

REGION-AWARE MASKING FOR ACCENT-ROBUST CROSS-LINGUAL TEXT-TO-SPEECH

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ABSTRACT

Accent leakage remains a critical challenge in cross-lingual zero-shot text-to-speech (TTS), where models inadvertently transfer both the speaker’s timbre and source-language accent into the target-language output. In mask-reconstruction TTS this is amplified by a train–inference mismatch: training reconstructs from same-language context, while inference conditions on cross-lingual context. We close this gap by concatenating utterances from two languages and placing the reconstruction mask relative to the language boundary, so training presents the cross-lingual prompting pattern the model encounters at inference. This requires no language IDs, accent labels, parallel same-speaker recordings, or architectural changes—only the mask geometry differs. In a listening study with native speakers, region-aware masking reduces perceived accent from 4.3 to 0.5 on a 0-5 scale against the unadapted baseline, and by 18-51% in seen regimes and 35-55% on unseen source languages against bilingual adaptation alone, with no measurable loss in intelligibility or naturalness. Comparing mask geometries shows that placement, not the amount masked, governs the balance between accent suppression and speaker preservation: masks confined to the reconstruction region suppress accent most but retain the reference speaker least reliably, whereas two-region masking achieves both. Mask geometry is thus a simple, effective lever for accent robustness.

Index Terms— cross-lingual speech synthesis, zero-shot text-to-speech, accent leakage, flow matching, speaker similarity

1. INTRODUCTION

Cross-lingual zero-shot text-to-speech (TTS) often exhibits accent leakage: although the synthesized speech preserves the reference speaker’s timbre, its target-language pronunciation commonly inherits the source-language accent. This issue arises in voice-preserving speech translation, media localization, and multilingual voice interfaces, where speaker characteristics should remain stable while pronunciation follows target-language norms. However, addressing this challenge through direct supervised learning is difficult, as parallel cross-lingual speech from the same speaker is scarce and expensive to collect at scale.

The problem stems from two related sources. First, speaker timbre is entangled with language-dependent pronunciation and prosody, allowing a model to encode them jointly. Prior work addresses this using language embeddings or tokens, disentangled speaker and linguistic representations, adversarial accent suppression, language-dependent generators, or inference-time model fusion [1, 2, 3, 4, 5, 6, 7]. While these methods highlight the need to separate speaker-related cues from pronunciation, they leave open a complementary question: can the training objective itself provide an effective separation signal without explicit language IDs, accent, or other token- or embedding-based conditioning?

Second, mask-reconstructing TTS introduces a fundamental train–inference mismatch. Voicebox [8], E2 TTS [9], and F5-TTS [10] learn speech infilling by masking part of an utterance and reconstructing it from text and surrounding audio. During standard training, both the masked target and its acoustic context come from the same language and utterance. During cross-lingual inference, however, target speech in one language must be generated using acoustic context from another. This mismatch encourages the model to copy language-dependent acoustic attributes—most notably, the source-language accent—from the reference audio rather than relying on the target text for pronunciation.

In this work, we propose a region-aware masking method for accent-robust cross-lingual TTS. Building on F5-TTS [10], we construct pseudo-bilingual examples by concatenating utterances from two distinct languages, optionally guided by speaker-embedding similarity. By reconstructing speech in one language while conditioning on the other as acoustic context, the model is exposed during training to the same cross-lingual prompting pattern encountered at inference time. This directly targets the train–inference mismatch underlying accent leakage, without explicit language IDs, accent conditioning, or architectural modifications.

To summarize, our main contributions are:

- **A region-aware cross-lingual adaptation strategy:** We construct pseudo-bilingual training pairs from speaker-disjoint monolingual data via region-aware masking, directly addressing the train–inference mismatch.
- **Conditioning-free accent control:** Our approach improves accent robustness without requiring explicit language IDs, accent embeddings, or model modifications.
- **A practical trade-off analysis:** We show that mask placement governs the trade-off between accent suppression and speaker-timbre preservation, identify two-region masking as the best overall setting, and demonstrate that the resulting gains generalize to unseen language combinations.

Audio demos are provided on the supplementary website.¹

2. RELATED WORK

Masked and cross-lingual generation. Prior work adapts masked reconstruction or codec-based language modeling for cross-lingual speech generation. Voicebox [8] introduced large-scale text-guided flow-matching infilling, and E2 TTS [9] and F5-TTS [10] simplified it to alignment-free zero-shot TTS. Cross-Lingual F5-TTS extends base F5-TTS toward language-agnostic voice cloning, dropping the reference transcript via forced alignment and speaking-rate prediction [14]. VALL-E established neural-codec language modeling for zero-shot TTS [11], and VALL-E X concatenates source- and

¹<https://doubleblindauthor.github.io/region-aware-masking-TTS-demo>

target-language phoneme prompts within a codec language model for cross-lingual synthesis [1]. SpeechPainter [12] and ERNIE-SAT [13] apply masked reconstruction directly to speech editing and synthesis. Separately, teacher-student augmentation synthesizes unseen speaker-language combinations for cross-lingual training [15].

Speaker, language, and accent separation. A separate line of work addresses accent leakage by separating speaker cues from linguistic content. Zhang *et al.* use adversarial training to separate the two and show that accented outputs can still receive higher speaker-similarity ratings [16], underscoring that similarity metrics alone cannot certify accent-free synthesis. YourTTS [17] and XTTS [18] extend zero-shot synthesis across languages. DSE-TTS separates a native-speaker representation for linguistic style from target-speaker timbre [2]; CrossSpeech++ uses separate language- and speaker-dependent generators [3]; AccentBox and CrossAccent-TTS learn explicit accent representations or suppression modules [4, 5]; X-Voice injects language identity at the text and frame levels [6]; PFluxTTS applies inference-time vector-field fusion [7]. Accent normalization has also been studied using non-parallel data and a separate discrete-token conversion model [20].

Both lines rely on masked or codec-based reconstruction with explicit language, accent, or speaker representations. We instead study how mask placement relative to the language boundary changes the training signal itself, improving accent robustness without language IDs, accent labels, or architectural changes.

3. METHOD

3.1. Masked flow-matching supervision

Our method builds on F5-TTS [10], a non-autoregressive diffusion-transformer TTS model trained for text-guided speech infilling. It uses optimal-transport conditional flow matching (OT-CFM) to learn a vector field between Gaussian noise and speech data [21, 22]. Let $x_1 \in \mathbb{R}^{T \times F}$ be a log-mel spectrogram and y its transcript. OT-CFM samples $x_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, and defines $\psi_t = (1 - t)x_0 + tx_1$ and $\dot{\psi}_t = x_1 - x_0$. Let $m \in \{0, 1\}^T$ indicate frames to reconstruct and $c = (1 - m) \odot x_1$ be the visible acoustic context. The masked CFM objective is

$$\mathcal{L}_{\text{MCFM}} = \mathbb{E}_{t, x_1 \sim q, x_0 \sim p_0, m \sim p_{\text{mask}}} \left\| m \odot \left(v_\theta(\psi_t, t, c, y) - \dot{\psi}_t \right) \right\|_2^2 \quad (1)$$

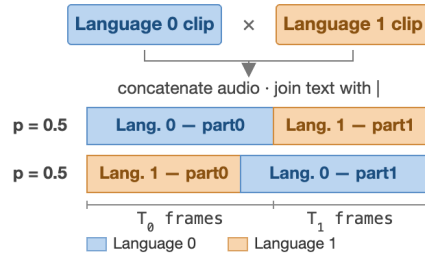
Here, v_θ is the time-dependent vector field predicted by F5-TTS, and q , p_0 , and p_{mask} denote the speech-data, Gaussian-noise, and training-mask distributions. Eq. (1) omits the normalization factor, as in prior work; in practice, we average over masked time-frequency bins. Standard F5-TTS masks one random contiguous span of length sT , where $s \sim \mathcal{U}[0.7, 1.0]$ [10].

3.2. Construction of concatenated bilingual examples

Let $\mathcal{D}_0$ and $\mathcal{D}_1$ denote monolingual corpora for Language 0 and Language 1. We sample one utterance from each corpus, concatenate their log-mel spectrograms, and join their character-token sequences with the in-vocabulary separator “|”. The concatenation order is sampled uniformly, exposing the model to both language directions. After ordering, we refer to the first and second positional regions as *part0* and *part1*; these names do not correspond to a fixed language. Let their durations be T_0 and T_1 mel frames.

The language boundary determines how the example is masked, but no language ID is passed to the model. The two utterances need not be semantically aligned or recorded from the same speaker. To preserve ordinary monolingual behavior, a fraction p_{single} of training examples remains monolingual and uses the standard F5-TTS mask;

(a) Pseudo-bilingual construction



(b) Region-aware mask policies

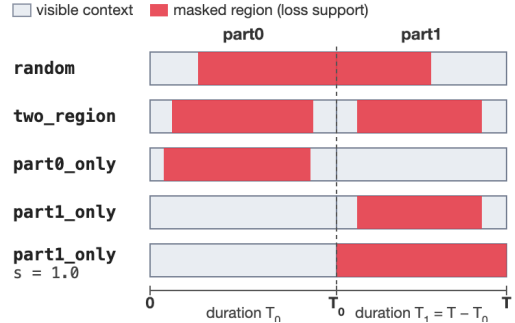


Fig. 1. Pseudo-bilingual example construction and region-aware masking. (a) Two utterances are concatenated into regions of durations T_0 and T_1 . (b) Gray denotes visible acoustic context and red denotes masked regions used in the flow-matching loss.

the remaining examples use the concatenated construction. At inference, the reference audio is prompted with its transcript and the target text joined by the same separator, and the model generates the target-language continuation.

3.3. Region-aware mask distributions

Our key idea is to place two languages in one training example, then mask one region so the model reconstructs it while receiving the other as acoustic context. This encourages transfer of language-independent cues such as timbre across the boundary while relying on the target text for pronunciation. The original random-span mask does not reliably create this condition, since it may cross the language boundary or leave target-region speech visible. We therefore define several position-aware masking strategies.

For an interval beginning at t_{start} and ending at t_{end} , let $\text{span}(t_{\text{start}}, t_{\text{end}}, s)$ denote a uniformly positioned contiguous mask of duration $\lfloor s(t_{\text{end}} - t_{\text{start}}) \rfloor$, with masked fraction s . With $T = T_0 + T_1$, we define four policies: *random* masks a span in $[0, T]$; *part0-only* masks a span in $[0, T_0]$; *part1-only* masks a span in $[T_0, T]$; and *two-region* independently masks one span in each region. Formally, these correspond to $\text{span}(0, T, s)$, $\text{span}(0, T_0, s)$, $\text{span}(T_0, T, s)$, and $\text{span}(0, T_0, s_0) \cup \text{span}(T_0, T, s_1)$, where $s, s_0, s_1 \stackrel{\text{iid}}{\sim} \mathcal{U}[0.7, 1.0]$ unless specified otherwise. The *uniform* policy samples one of the four with equal probability at each optimization step. Because concatenation order is randomized, *part0-only* and *part1-only* are symmetric over training.

The most inference-aligned policy is *part1-only*. Under the default span-fraction distribution, part of the second region remains visible; setting $s = 1$ masks it completely, requiring the model to reconstruct it from its transcript and cross-language acoustic context. With order randomized, either language can occupy the reconstruction region. This signal is intended to encourage, rather than guarantee, separation of timbre from language-dependent pronunciation.

3.4. Speaker-similarity-selected pseudo-pairs

Region-aware masking does not require any cross-lingual speaker matching: the simplest setting concatenates utterances from the two languages at random. Random concatenation, however, can pair very different voices, weakening cross-region acoustic context for speaker timbre. We therefore also study a pseudo-pairing strategy based on speaker similarity.

For each utterance in an anchor language, we retrieve the top K candidates from a candidate language by cosine similarity between normalized pretrained speaker embeddings and sample one candidate during training. This raises similarity in speaker characteristics only; it does not guarantee that the paired utterances share the same speaker. We therefore refer to them as *pseudo-pairs*.

To quantify how strongly an embedding model separates retrieved from random pairs, we define

$$\rho = \frac{\mathbb{E}_i[\cos(e_i, e_{NN_1(i)})]}{\mathbb{E}_{i,j}[\cos(e_i, e_j)]} \quad (2)$$

where e_i is an anchor-language embedding, e_j a randomly selected candidate-language embedding, and $e_{NN_1(i)}$ the nearest candidate-language embedding to e_i . A value near one indicates little separation between retrieved and random pairs under that encoder. This ratio is descriptive rather than a direct measure of pair quality: speaker embeddings may also encode channel, language, accent, or other attributes, and ρ evaluates the nearest candidate although training samples from the top K .

4. EXPERIMENTAL SETUP

Datasets. We instantiate Language 0 and Language 1 as English and Spanish. Both languages offer large, diverse corpora, and their phonological and prosodic differences provide a meaningful test of cross-lingual accent transfer. We construct pseudo-bilingual training examples from an internal corpus of approximately 21.2M English and 9.3M Spanish utterances. Each utterance is a single-speaker audio clip with durations between 2 and 16 seconds. The two language subsets are speaker-disjoint, and no same-speaker bilingual recordings are used.

For evaluation, we use an internal ES→EN test set of 20 verified cross-lingual speech-text pairs, with Spanish reference speech and English target text. To assess accent robustness beyond the training language pair and domain, we also build a test set from FLEURS [25], which provides semantically parallel speech and text across languages. We select 30 sentences and pair their Spanish, French, and Portuguese recordings with the corresponding English target text, yielding 90 cross-lingual items across three source languages. All paired utterances preserve the same semantic content.

Cross-lingual speaker-pair retrieval. For the optional pairing strategy, we precompute cross-language nearest-neighbor tables with FAISS [26], avoiding online similarity search during training. English utterances serve as anchors, Spanish utterances as candidates. For each anchor, we retrieve the top $K = 10$ candidates by cosine similarity between normalized speaker embeddings and sample one uniformly from this set, balancing speaker similarity with pair diversity. After pairing, the concatenation order during training is sampled uniformly. Table 1 reports the top-1 retrieval contrast from Eq. (2) over 1,000 anchors. ECAPA-TDNN yields a substantially larger retrieved-versus-random contrast than x-vector, indicating stronger discrimination between retrieved and random pairs and motivating its use in the optional speaker-pairing experiments.

Model setup. We adopt two measures to avoid degrading monolingual generation. First, all systems are initialized from the same

Table 1. Top-1 speaker-pair retrieval statistics.

Encoder	Dim.	NN cosine	Random cosine	ρ
x-vector [23]	512	0.982	0.923	1.06
ECAPA-TDNN [24]	192	0.762	0.322	2.37

Table 2. Systems. $U = \mathcal{U}[.7, 1]$.

System	Mask (s)	Pairing
B	random (U)	None (single-language utterance)
M1	random (U)	Random
M2	two_region (U)	Random
M3	part1_only (U)	Random
M4	part1_only (1)	Random
M5	uniform (U)	Random
M5+P	uniform (U)	ECAPA-TDNN
M2+P	two_region (U)	ECAPA-TDNN

pretrained multilingual F5-TTS checkpoint, rather than training from scratch: we evaluate it directly as baseline B and adapt the rest from it. Second, alongside concatenated bilingual examples, we retain a small proportion of monolingual examples during adaptation to mitigate catastrophic forgetting.

We set $p_{\text{single}} = 0.2$, so 80% of adaptation examples are concatenated and 20% remain monolingual. During training, we use AdamW with a learning rate of 10^{-4} and a 4,000-update warmup. All systems are compared at the same training budget of 2.04 epochs on the concatenated mixture, so differences reflect the method rather than training length.

We evaluate the masking methods summarized in Table 2. In the Mask column, *uniform* denotes a mixture of all four masking policies: At each training step, one of the policies defined in Section 3.3 is sampled with equal probability. U indicates that the corresponding mask fraction is sampled from $\mathcal{U}[0.7, 1.0]$. M+P denotes the corresponding masking policy with the optional ECAPA-TDNN-based speaker-pair retrieval.

Evaluation. We report three objective measures, along with a subjective listening test. Speaker similarity (SIM) is computed using both WavLM [27] and ECAPA-TDNN [24]. Since ECAPA-TDNN is also used for pseudo-pair retrieval, WavLM provides an independent check against potential bias from pair selection. Intelligibility is measured by word error rate (WER) using Whisper large-v3 [28], and naturalness is estimated with UTMOS [29]. We treat UTMOS as a proxy for naturalness and overall generation quality.

To directly assess accent robustness, we conduct a human listening test. Each page presents a non-English reference utterance, the English target text, and the generated English speech. Twenty English native speakers rate each clip on two criteria. The first rates the perceived strength of the source-language accent on a 0–5 scale: 0 none, 1 very slight, 2 mild, 3 moderate, 4 strong, and 5 very strong; lower is better. The second asks whether the speaker timbre is noticeably changed relative to the reference, which we report as the fraction of clips flagged.

5. RESULTS AND DISCUSSION

We report results in three configurations that cross two axes of shift: source language and audio domain. **Internal ES→EN** is in-domain on both. **FLEURS ES→EN** keeps the language pair and switches to out-of-domain (OOD) audio. **FLEURS FR/PT→EN** shifts both at once, with French and Portuguese unseen during adaptation and retrieval. The first two test the adapted model’s language-pair performance across domains; the third evaluates generalization to unseen source languages. In the listening study, 20 raters score each clip for accent on a 0–5 scale and judge whether its timbre is preserved.

Inter-rater agreement is high for accent (Krippendorff’s $\alpha = 0.98$) and moderate for timbre ($\alpha = 0.53$), so we interpret the latter mainly at the level of large between-policy differences.

Region-aware masking consistently reduces accent leakage.

The unadapted baseline **B** exhibits strong accent leakage across all three regimes, with accent scores of 4.21–4.37, showing that cross-lingual prompting largely transfers the source-language accent together with the reference speaker’s timbre. Adapting the pre-trained multilingual checkpoint on bilingual concatenations already mitigates this effect substantially: **M1** reduces the accent score to 0.45 on the internal set, 0.69 on FLEURS ES→EN, and 1.05 on FLEURS FR/PT→EN. All region-aware masking variants further improve over **M1**.

Compared to **M1**, region-aware masking reduces perceived accent by 18–51% in the seen regimes and 35–55% on unseen source languages. These gains are significant in all settings except **M5** on the 20-item internal set. The largest gains appear in the hardest regime, **FLEURS FR/PT→EN**, where adaptation provides no direct support for the source languages: the accent score drops from 1.05 for **M1** to 0.69 for **M2**, 0.60 for **M5**, and 0.48–0.54 for **M3/M4**. The trend holds in-domain and out-of-domain, suggesting that the gain comes from the training strategy, not the recording domain.

Accent suppression and speaker preservation trade off. The gains from region-aware masking come with a clear trade-off, but it is driven by where the mask falls, not by how much it covers: **random** and **two-region** both mask 85% of the example in expectation, yet spreading the mask across both regions lowers accent and raises **Joint**. The **part1-only** variants mask only 42.5% in expectation, yet preserve timbre least reliably. On unseen languages, accent improves from 0.69 with **M2** to 0.48 with **M3**, but the share of clips that deviate from the reference speaker’s timbre rises sharply from 15.1% to 42.1%. SIM exhibits the same trend: the part1-only variants show the largest similarity drop. In several such cases the change is perceptually obvious, indicating that these failures are substantial rather than marginal.

This trade-off makes accent score alone an inadequate selection criterion. If systems are ranked only by accent suppression, these part1-only variants appear best; but they also fail most often to preserve the reference speaker’s timbre, making them unsuitable for voice-preserving applications. To capture both requirements, we further report **Joint**: the fraction of clips with both accent score ≤ 1 and preserved speaker timbre. This threshold corresponds to no accent or only a very slight accent. Under this criterion the ranking reverses: on unseen languages, **M2**, **M2+P**, **M5** and **M5+P** are tied-best on Joint (66–70%), while **M3** and **M4** drop to 51.3% and 42.5%. We therefore recommend **two-region** masking as the default.

Intelligibility and naturalness are largely preserved. Accent-robustness gains bring no measurable loss in intelligibility or naturalness. Across the adapted systems, WER and UTMOS differences are generally not statistically resolvable in the seen configurations; UTMOS separates the unadapted baseline in both, while WER does so only on FLEURS. This suggests the proposed adaptation mainly changes accent and speaker-preservation behavior while leaving overall generation quality largely intact. The only notable exception is **M5** on FLEURS FR/PT→EN, which improves over **M1** in WER; no other masking variant shows this pattern.

Speaker-similarity pairing is an optional extension that mainly improves SIM under aggressive masking. We include pseudo-pair retrieval because purely random cross-lingual concatenation can pair very different voices. Its effect is smaller and more selective than that of masking itself. Under **two-region** masking, some same-speaker acoustic context remains visible, so retrieval

Table 3. Objective and subjective results by evaluation regime. Joint is the share of clips with accent ≤ 1 and preserved speaker timbre. SIM reports WavLM / ECAPA-TDNN. Gray rows (**B**, **M1**) are reference points; bold marks the tied-best proposed systems (paired bootstrap, $p > 0.05$). WER is unbolded in-domain, where no adapted systems are statistically distinguishable.

System	Accent↓	Timbre chg. %↓	Joint %↑	SIM↑ (WavLM / ECAPA)	WER↓	UTMOS↑
<i>In-domain — Internal ES→EN</i>						
B	4.28	1.5	1.3	0.953 / 0.701	0.051	2.88
M1	0.45	10.8	79.1	0.673 / 0.313	0.042	3.77
M2	0.32	14.7	80.2	0.645 / 0.249	0.060	3.85
M3	0.22	45.3	52.3	0.520 / 0.079	0.024	4.01
M4	0.26	51.0	45.2	0.478 / 0.066	0.039	3.93
M5	0.37	21.3	71.1	0.614 / 0.198	0.036	3.86
M5+P	0.23	26.5	70.6	0.578 / 0.171	0.054	3.91
M2+P	0.36	11.4	82.4	0.653 / 0.261	0.042	3.95
<i>OOD, seen source language — FLEURS ES→EN</i>						
B	4.37	1.4	0.7	0.951 / 0.717	0.070	3.38
M1	0.69	7.4	75.3	0.722 / 0.406	0.049	3.97
M2	0.50	12.7	75.9	0.671 / 0.270	0.049	3.90
M3	0.45	55.0	40.1	0.507 / 0.073	0.037	3.88
M4	0.36	73.5	23.4	0.427 / 0.069	0.037	4.00
M5	0.45	29.2	62.4	0.623 / 0.218	0.058	3.86
M5+P	0.40	25.2	66.9	0.603 / 0.224	0.058	3.95
M2+P	0.52	10.0	77.4	0.699 / 0.316	0.055	4.04
<i>OOD, unseen source languages — FLEURS FR/PT→EN</i>						
B	4.21	1.5	3.1	0.960 / 0.710	0.122	3.20
M1	1.05	10.0	64.5	0.776 / 0.358	0.089	3.81
M2	0.69	15.1	69.6	0.704 / 0.234	0.066	3.94
M3	0.48	42.1	51.3	0.558 / 0.072	0.064	3.89
M4	0.54	52.2	42.5	0.535 / 0.053	0.066	3.85
M5	0.60	22.0	65.7	0.663 / 0.168	0.035	3.93
M5+P	0.62	16.2	69.4	0.694 / 0.241	0.055	3.87
M2+P	0.67	16.0	69.7	0.718 / 0.277	0.038	3.92

yields only modest SIM gains (+0.046 on FLEURS ES and +0.043 on unseen languages). Under **uniform** masking, the target region can be fully masked, leaving no same-speaker evidence; the retrieved pair then becomes the only speaker anchor and yields the largest gain we observe (+0.073 on unseen languages), while it is small on FLEURS ES (+0.006) and slightly negative in-domain. Because ECAPA-TDNN is used both for retrieval and as one similarity metric, we also report WavLM-based SIM as an independent check. The gains persist there at smaller magnitude. However, these objective improvements do not translate into a clearly resolved gain in human timbre judgments, and pairing does not materially affect accent.

We therefore view pairing as optional: it can improve similarity when masking removes most speaker evidence, but the main gains in accent robustness come from region-aware masking. More broadly, SIM should be interpreted carefully in cross-lingual accent evaluation, since higher SIM can also reflect stronger source-accent copying. This is most evident in baseline **B**, which achieves the highest SIM while also exhibiting the strongest accent leakage. Taken together, these results identify **two-region** masking as the best trade-off between accent robustness and speaker-timbre preservation.

6. CONCLUSION

We introduced a region-aware masking strategy for accent-robust cross-lingual TTS. By reconstructing speech in one language from acoustic context in another, the method better aligns masked training with cross-lingual inference, without architectural changes, explicit language or accent conditioning, or same-speaker bilingual recordings. Experiments show that this change to the training signal consistently reduces perceived accent, generalizes to unseen source languages, and reveals a clear trade-off between accent suppression and speaker-timbre preservation. Mask geometry thus provides an effective lever for controlling accent robustness.

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